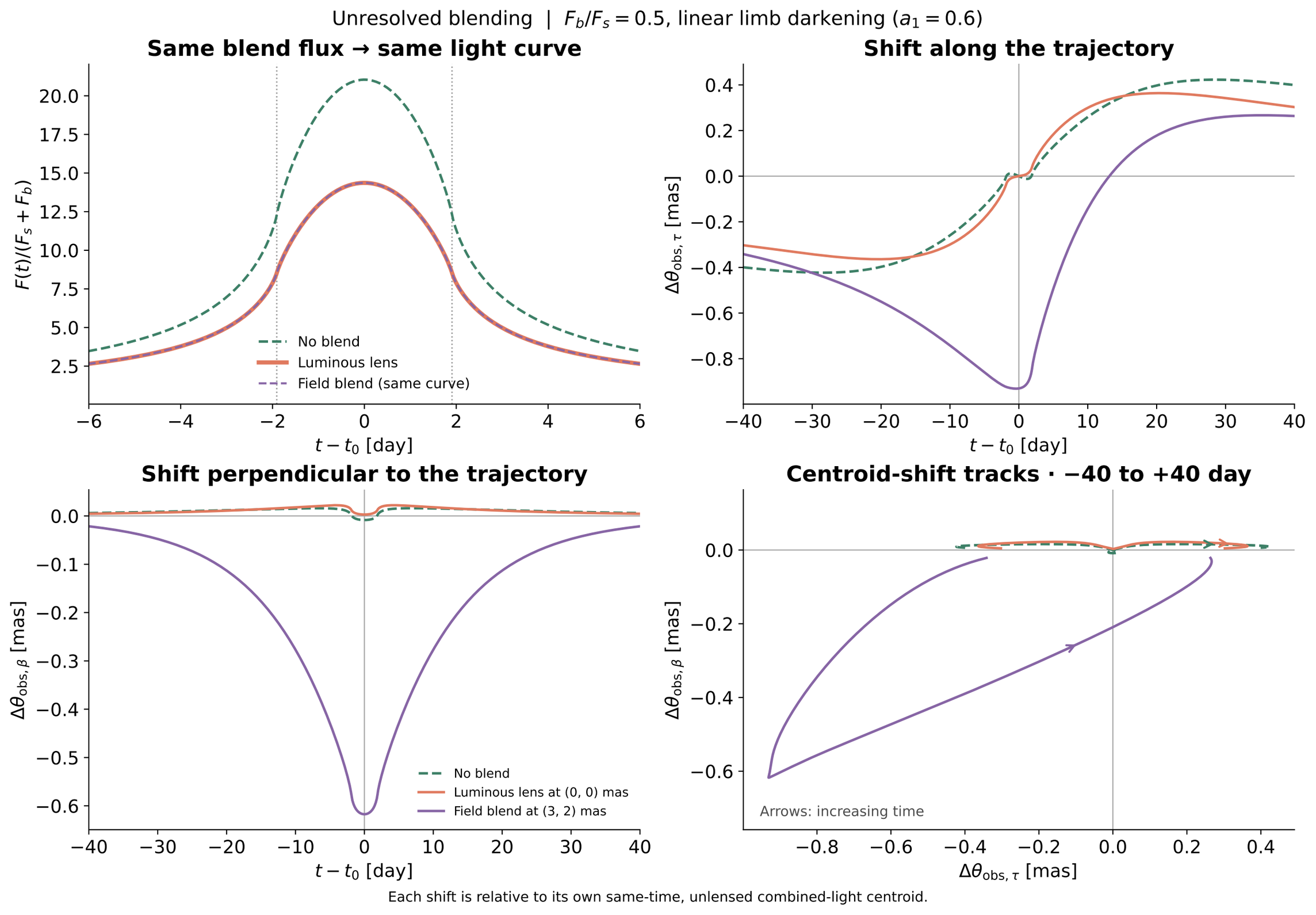
Unresolved blending | $F_b/F_s = 0.5$, linear limb darkening ($a_1 = 0.6$)
Same blend flux → same light curve
$F(t)/(F_s + F_b)$
No blend
Luminous lens
Field blend (same curve)
$t - t_0$ [day]
Shift along the trajectory
$\Delta\theta_{\mathrm{obs},\tau}$ [mas]
$t - t_0$ [day]
Shift perpendicular to the trajectory
$\Delta\theta_{\mathrm{obs},\beta}$ [mas]
No blend
Luminous lens at (0, 0) mas
Field blend at (3, 2) mas
$t - t_0$ [day]
Centroid-shift tracks · −40 to +40 day
$\Delta\theta_{\mathrm{obs},\beta}$ [mas]
Arrows: increasing time
$\Delta\theta_{\mathrm{obs},\tau}$ [mas]
Each shift is relative to its own same-time, unlensed combined-light centroid.